

# Summary of Space Geometry Laws

## 1. Coordinate System in 3D (النظام الإحداثي)

- Distance between two points A  $(x_1, y_1, z_1)$  and B  $(x_2, y_2, z_2)$ :  
 $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$
- Midpoint of AB:  $M = (\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2})$
- Equation of a Sphere (معادلة الكرة):  
 $(x - L)^2 + (y - K)^2 + (z - N)^2 = r^2$  (Where  $(L, K, N)$  is the center and  $r$  is the radius)

## 2. Vectors in Space (المتجهات في الفراغ)

- Magnitude of Vector  $A = (A_x, A_y, A_z)$ :

$$\|A\| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

- Unit Vector (متجه الوحدة):  $uA = \frac{A}{\|A\|}$
- Direction Cosines (زوايا الاتجاه):

$$\cos^2 \theta_x + \cos^2 \theta_y + \cos^2 \theta_z = 1$$

- Vector Components:  $A = \|A\|(\cos \theta_x, \cos \theta_y, \cos \theta_z)$

## 3. Scalar Product - Dot Product (الضرب القياسي)

- Formula:**  $A \cdot B = \|A\| \|B\| \cos \theta$
- Algebraic form:**  $A \cdot B = A_x B_x + A_y B_y + A_z B_z$
- Angle between two vectors:  $\cos \theta = \frac{A \cdot B}{\|A\| \|B\|}$
- Work Done (الشغل):**  $W = F \cdot S = \|F\| \|S\| \cos \theta$
- Perpendicular Vectors (التعامد):**  $A \cdot B = 0$

## 4. Vector Product - Cross Product (الضرب الاتجاهي)

- Formula:**  $A \times B = (\|A\| \|B\| \sin \theta) n$
- Using Determinants:  $A \times B = \begin{vmatrix} i & j & k \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$
- Area of Parallelogram (متوازي الأضلاع):**  $Area = \|A \times B\|$
- Area of Triangle (المثلث):**  $Area = \frac{1}{2} \|A \times B\|$

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- **Parallel Vectors (التوازي):**  $A \times B = 0$  or  $\frac{A_x}{B_x} = \frac{A_y}{B_y} = \frac{A_z}{B_z}$

## 5. Vector Projections (المساقط)

- **Scalar Component (المركبة الجبرية)** of A in direction of B =  $\frac{A \cdot B}{\|B\|}$  or  $\|A\| \cos \theta$
- **Vector Component (المركبة الاتجاهية)** of A in direction of B =  $\frac{A \cdot B}{\|B\|^2} \cdot \frac{B}{\|B\|}$

## 6. Straight Line in Space (معادلة المستقيم)

- **Vector Form:**  $r = A + kd$
- **Parametric Form:**  $x = x_1 + ka, y = y_1 + kb, z = z_1 + kc$
- **Cartesian Form:**  $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$

## 7. Angle between Two Lines (الزاوية بين مستقيمين)

For two lines with direction vectors  $d_1$  and  $d_2$ :

- **Main Formula:**  $\cos \theta = \frac{|d_1 \cdot d_2|}{\|d_1\| \|d_2\|}$
- **Using Direction Cosines:**  $\cos \theta = |L_1 L_2 + M_1 M_2 + N_1 N_2|$

## 8. Positions of Two Lines (أوضاع مستقيمين)

For lines  $L_1$  and  $L_2$ :

- **Angle between two lines:**  $\cos \theta = \frac{|L_1 \cdot L_2|}{\|L_1\| \|L_2\|}$
- **Parallel Lines ( $L_1 \parallel L_2$ ):** If  $d_1 = kd_2$  or  $d_1 \times d_2 = 0$  or  $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
- **Perpendicular Lines ( $L_1 \perp L_2$ ):** If  $d_1 \cdot d_2 = 0$ , which means  $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$
- **Intersecting Lines (متقاطعان):** If they share a common point.
- **Skew Lines (متخالفان):** If they are neither parallel nor intersecting (not in the same plane).

## 9. Distance from a Point to a Line (البعد بين نقطة ومستقيم)

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To find the length of the perpendicular from point P to a line passing through point A with direction vector d:

- **Length (L):**  $L = \frac{\|AP \times d\|}{\|d\|}$
- **Vector AP:**  $P - A$

### 10. Intersection of Two Planes (خط تقاطع مستويين)

To find the line of intersection:

- **Direction Vector (d):**  $d = n_1 \times n_2$
- **Finding a Point:** Solve the two plane equations together by assuming a value for one variable (e.g.,  $z = 0$ ).

### Summary of Important Notes

- **Line // Plane:**  $n \cdot d = 0$  (Normal vector is perpendicular to direction vector).
- **Line  $\perp$  Plane:**  $n \parallel d$  (Normal vector is parallel to direction vector).
- **Angle between Line and Plane:**  $\sin \theta = \frac{|n \cdot d|}{\|n\| \|d\|}$

### 11. Equation of a Plane (معادلة المستوى)

To find the equation, we need a point P ( $x_1, y_1, z_1$ ) and a normal vector  $n = (a, b, c)$

- **Vector Form (الصورة المتجهة):**  $n \cdot r = n \cdot P$
- **Standard/General Form (الصورة العامة):**  
 $ax + by + cz + d = 0$
- **Point-Normal Form (الصورة القياسية):**  
 $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$

### 12. Length of Perpendicular (طول العمود)

The length L from a point ( $x_1, y_1, z_1$ ) to the plane  $ax + by + cz + d = 0$ :

$$L = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

### 13. Relationship between Two Planes (العلاقة بين مستويين)

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For planes with normal vectors  $n_1$  and  $n_2$ :

- **Parallel Planes (التوازي):**  $n_1 // n_2$  or  $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
- **Perpendicular Planes (التعامد):**  $n_1 \cdot n_2 = 0$
- Angle between two planes:  $\cos \theta = \frac{|n_1 \cdot n_2|}{\|n_1\| \|n_2\|}$

## 14. Relationship between Line and Plane (العلاقة بين مستقيم ومستوى)

- Angle between Line ( $d$ ) and Plane ( $n$ ):  $\sin \theta = \frac{|n \cdot d|}{\|n\| \|d\|}$
- **Line parallel to Plane:**  $n \cdot d = 0$
- **Line perpendicular to Plane:**  $n // d$

